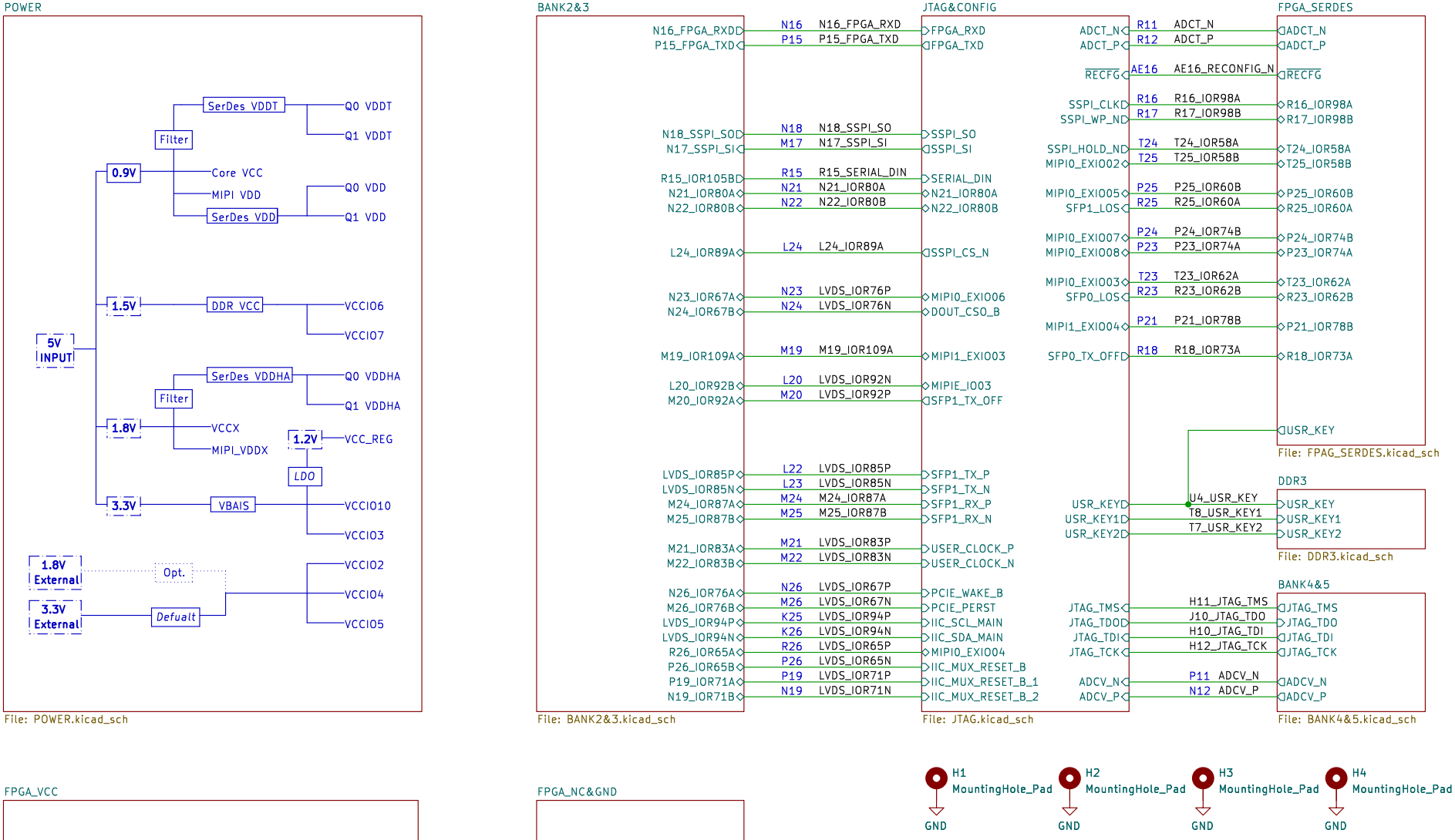


A Compact SOM for Arora V

Shenzhen Sipeed Technology Co,Ltd.

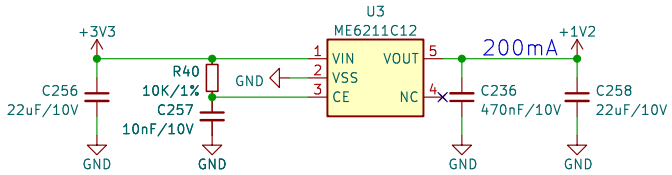
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File: TANG_MEGA-138K-Pro_4072.kicad_sch

Title: TANG_MEGA-138K-Pro_4072_Schematics

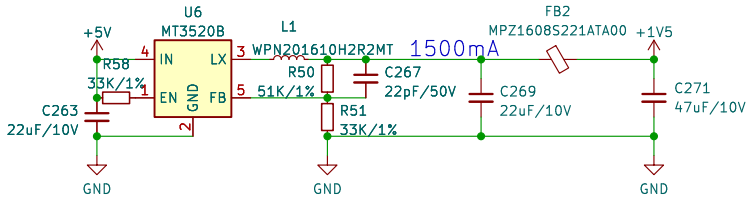
Size: A4	Date: 2024-03-25	**Rev: 1.10**
KiCad E.D.A. kicad 7.0.10		Id: 1/9

1.2V LDO



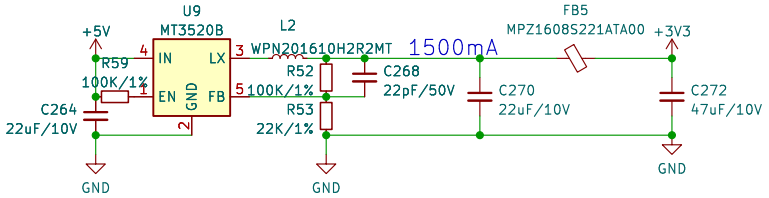
1V5 DC-DC

Vout=0.6*(1+51K/33K)≈1.53V

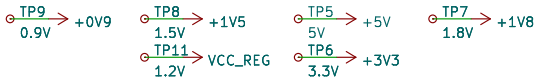


3V3 DC-DC

Vout=0.6*(1+100K/22K)≈3.33V

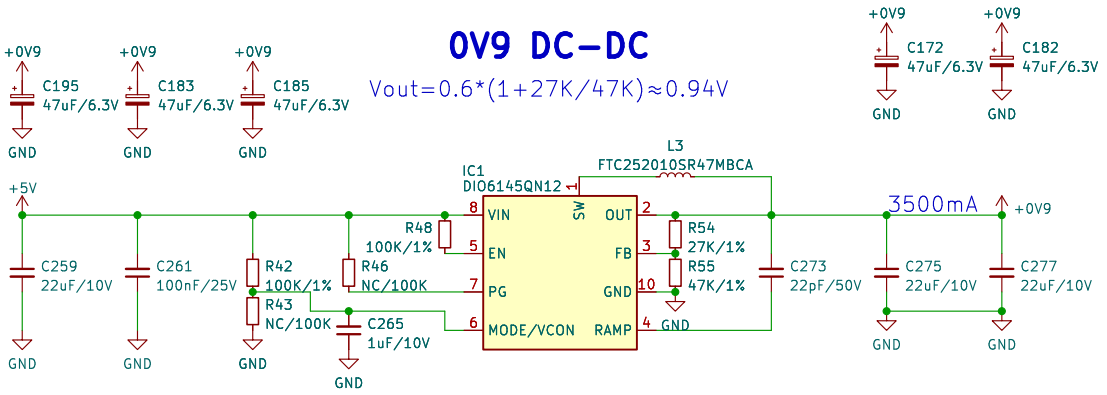


TEST POINTS



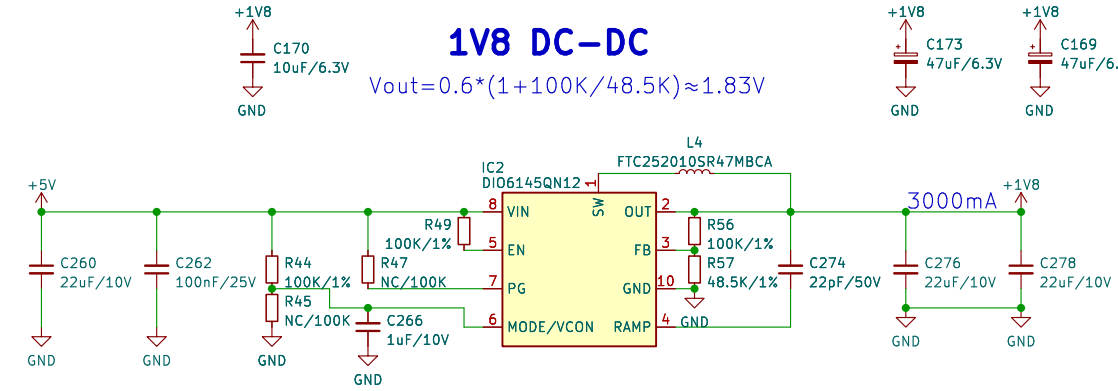
0V9 DC-DC

Vout=0.6*(1+27K/47K)≈0.94V



1V8 DC-DC

Vout=0.6*(1+100K/48.5K)≈1.83V



A Compact SOM for Arora V

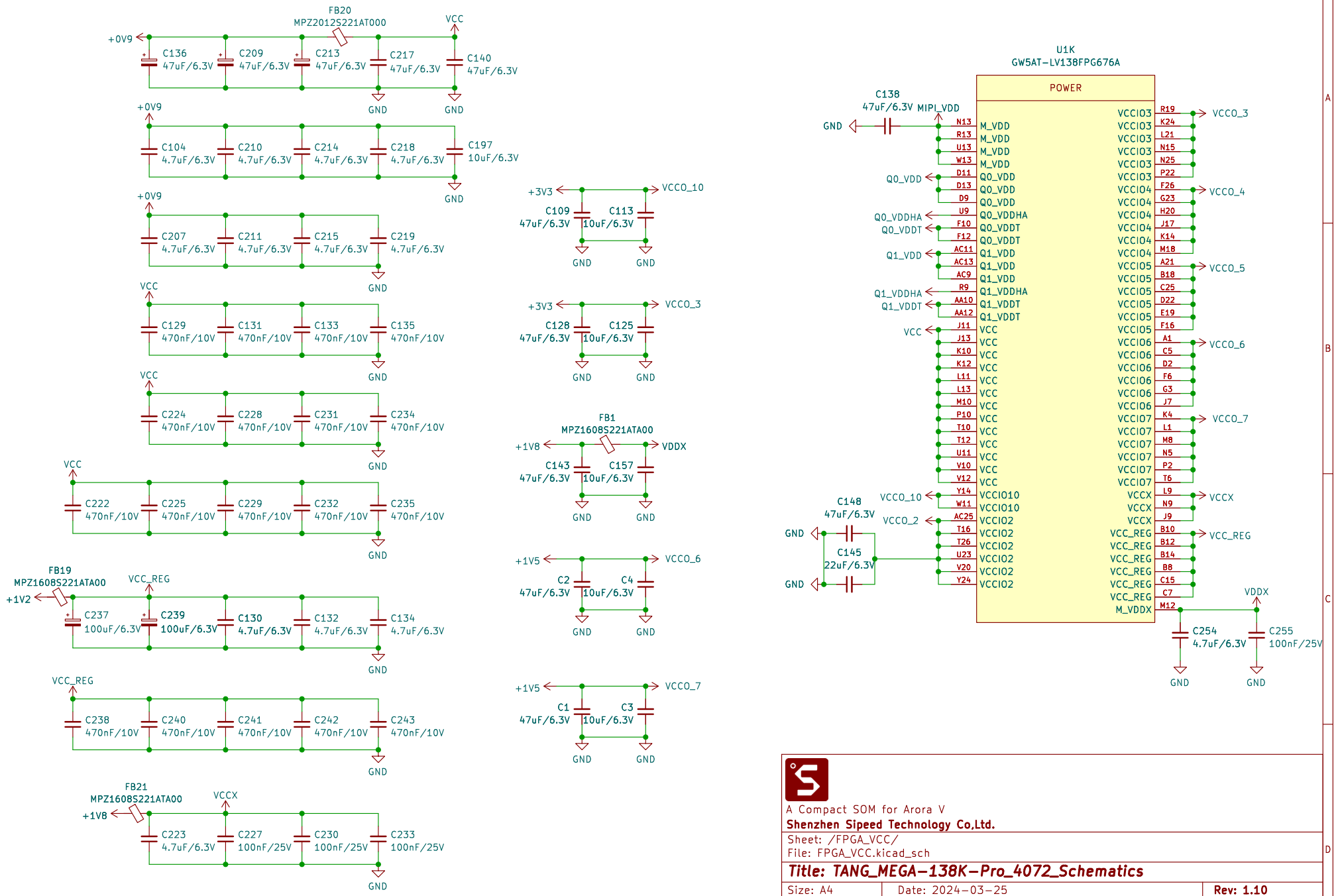
Shenzhen Sipeed Technology Co,Ltd.

Sheet: /POWER/

File: POWER.kicad_sch

Title: TANG_MEGA-138K-Pro_4072_Schematics

Size: A4	Date: 2024-03-25	**Rev: 1.10**
KiCad E.D.A. kicad 7.0.10		Id: 2/9


FB20
MPZ2012S221AT000
+0V9
VCC
C136 47uF/6.3V
C209 47uF/6.3V
C213 47uF/6.3V
C217 47uF/6.3V
C140 47uF/6.3V
GND
C104 4.7uF/6.3V
C210 4.7uF/6.3V
C214 4.7uF/6.3V
C218 4.7uF/6.3V
C197 10uF/6.3V
C207 4.7uF/6.3V
C211 4.7uF/6.3V
C215 4.7uF/6.3V
C219 4.7uF/6.3V
C129 470nF/10V
C131 470nF/10V
C133 470nF/10V
C135 470nF/10V
C224 470nF/10V
C228 470nF/10V
C231 470nF/10V
C234 470nF/10V
C222 470nF/10V
C225 470nF/10V
C229 470nF/10V
C232 470nF/10V
C235 470nF/10V
FB19
MPZ1608S221ATA00
+1V2
VCC_REG
C237 100uF/6.3V
C239 100uF/6.3V
C130 4.7uF/6.3V
C132 4.7uF/6.3V
C134 4.7uF/6.3V
C238 470nF/10V
C240 470nF/10V
C241 470nF/10V
C242 470nF/10V
C243 470nF/10V
FB21
MPZ1608S221ATA00
+1V8
VCCX
C223 4.7uF/6.3V
C227 100nF/25V
C230 100nF/25V
C233 100nF/25V
+3V3
VCCO_10
C109 47uF/6.3V
C113 10uF/6.3V
VCCO_3
C128 47uF/6.3V
C125 10uF/6.3V
FB1
MPZ1608S221ATA00
VDDX
C143 47uF/6.3V
C157 10uF/6.3V
+1V5
VCCO_6
C2 47uF/6.3V
C4 10uF/6.3V
VCCO_7
C1 47uF/6.3V
C3 10uF/6.3V
U1K
GW5AT-LV138FPG676A
POWER
C138 47uF/6.3V
MIPI_VDD
Q0_VDD
Q0_VDDHA
Q0_VDDT
Q1_VDD
Q1_VDDHA
Q1_VDDT
VCC
VCCO_10
VCCO_2
C148 47uF/6.3V
C145 22uF/6.3V
M_VDD
Q0_VDD
Q0_VDDHA
Q0_VDDT
Q1_VDD
Q1_VDDHA
Q1_VDDT
VCC
VCCIO10
VCCIO2
VCCIO3
VCCIO4
VCCIO5
VCCIO6
VCCIO7
VCCX
VCC_REG
M_VDDX
C254 4.7uF/6.3V
C255 100nF/25V
A Compact SOM for Arora V
Shenzhen Sipeed Technology Co,Ltd.
Sheet: /FPGA_VCC/
File: FPGA_VCC.kicad_sch
Title: TANG_MEGA-138K-Pro_4072_Schematics
Size: A4
Date: 2024-03-25
Rev: 1.10
KiCad E.D.A. kicad 7.0.10
Id: 3/9



A Compact SOM for Arora V

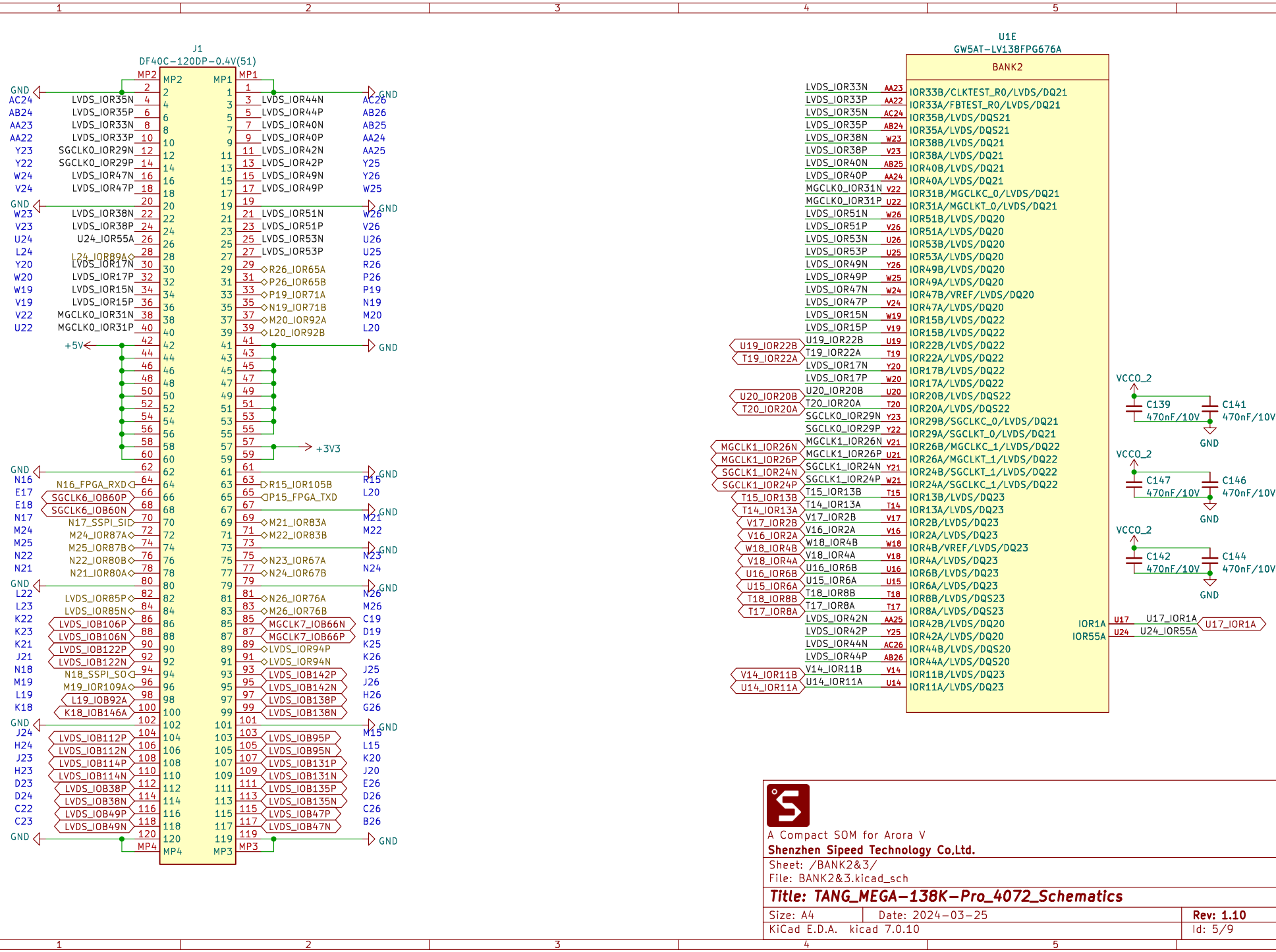
Shenzhen Sipeed Technology Co,Ltd.

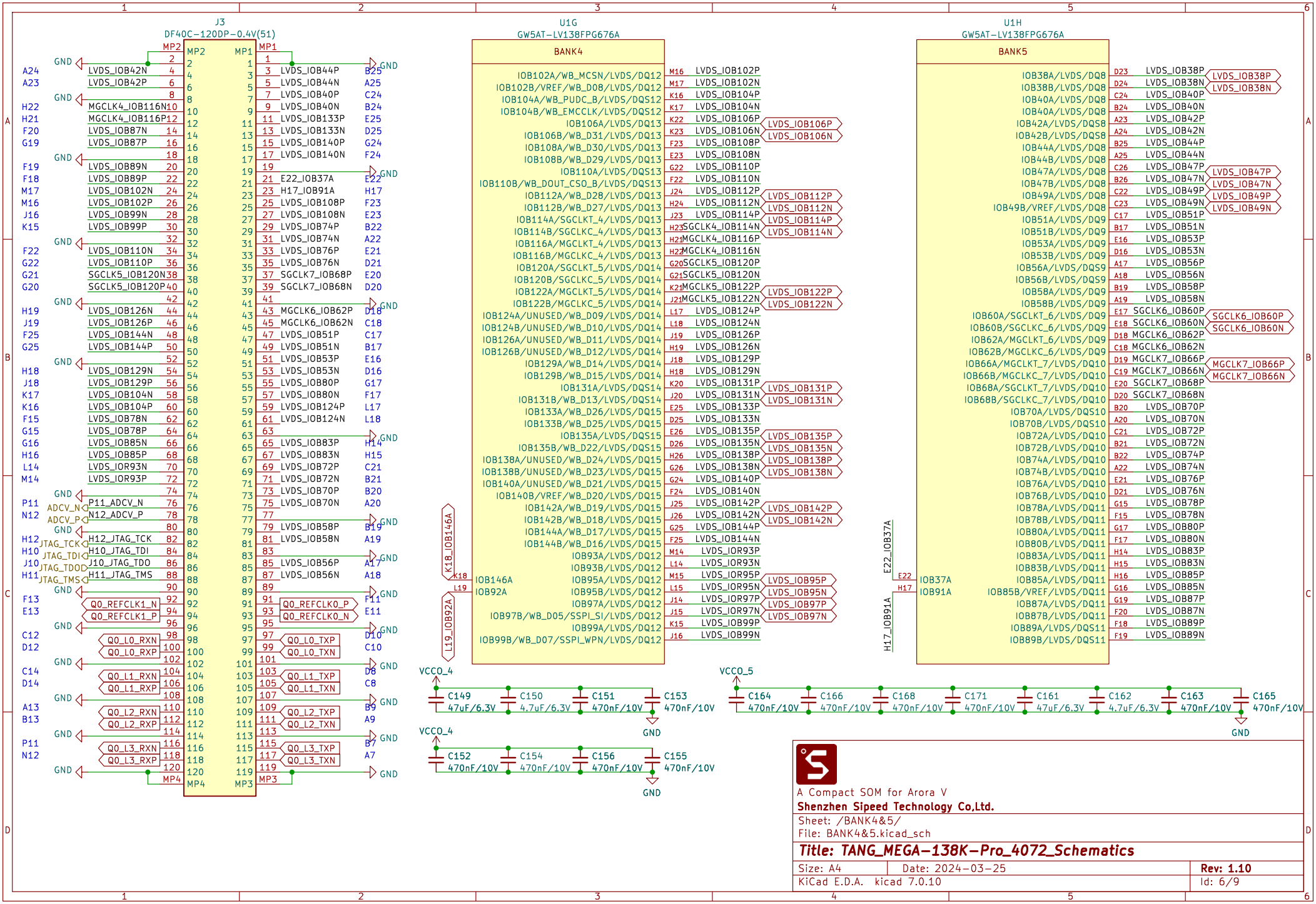
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File: FPGA_NC&GND.kicad_sch

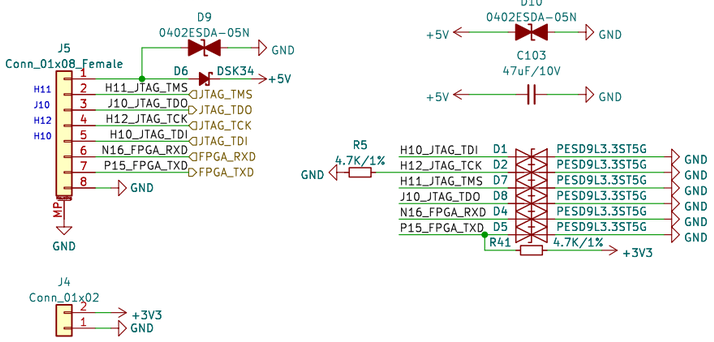
Title: TANG_MEGA-138K-Pro_4072_Schematics

Size: A4	Date: 2024-03-25	**Rev: 1.10**
KiCad E.D.A. kicad 7.0.10		Id: 4/9

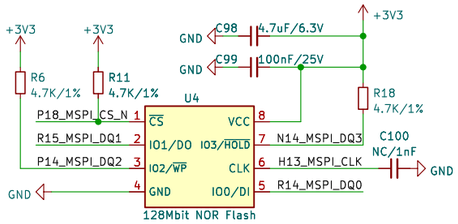

J1
DF40C-120DP-0.4V(51)
U1E
GW5AT-LV138FPG676A
BANK2
IOR1A
IOR55A
C139 470nF/10V
C141 470nF/10V
C147 470nF/10V
C146 470nF/10V
C142 470nF/10V
C144 470nF/10V
VCCO_2
GND
+5V
+3V3
A Compact SOM for Arora V
Shenzhen Sipeed Technology Co,Ltd.
Sheet: /BANK2&3/
File: BANK2&3.kicad_sch
Title: TANG_MEGA-138K-Pro_4072_Schematics
Size: A4
Date: 2024-03-25
Rev: 1.10
KiCad E.D.A. kicad 7.0.10
Id: 5/9


J3
DF40C-120DP-0.4V(51)
U1G
GW5AT-LV138FPG676A
BANK4
U1H
GW5AT-LV138FPG676A
BANK5
VCCO_4
VCCO_5
C149 47uF/6.3V
C150 4.7uF/6.3V
C151 470nF/10V
C153 470nF/10V
C152 470nF/10V
C154 470nF/10V
C156 470nF/10V
C155 470nF/10V
C164 470nF/10V
C166 470nF/10V
C168 470nF/10V
C171 470nF/10V
C161 47uF/6.3V
C162 4.7uF/6.3V
C163 470nF/10V
C165 470nF/10V
GND
A Compact SOM for Arora V
Shenzhen Sipeed Technology Co,Ltd.
Sheet: /BANK4&5/
File: BANK4&5.kicad_sch
Title: TANG_MEGA-138K-Pro_4072_Schematics
Size: A4
Date: 2024-03-25
Rev: 1.10
KiCad E.D.A. kicad 7.0.10
Id: 6/9

JTAG & UART

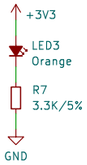


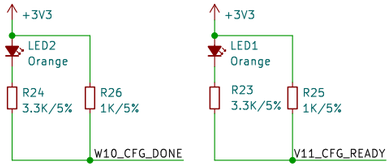
MSPI_FLASH



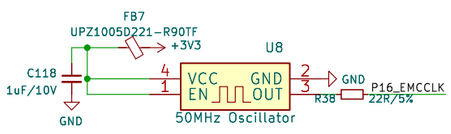


LEDs

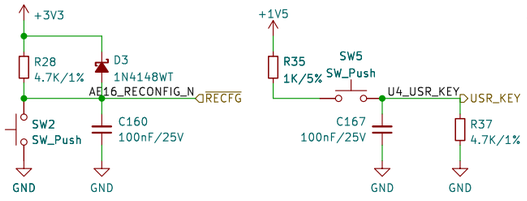




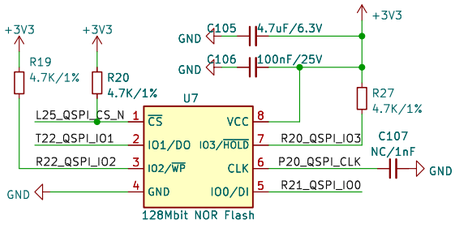
EMCCLK OSC.



BUTTONs



QSPI_NAND



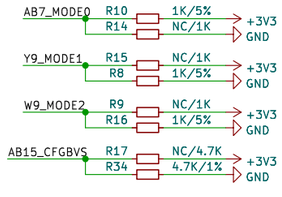
ADC INPUT



TESTPOINT



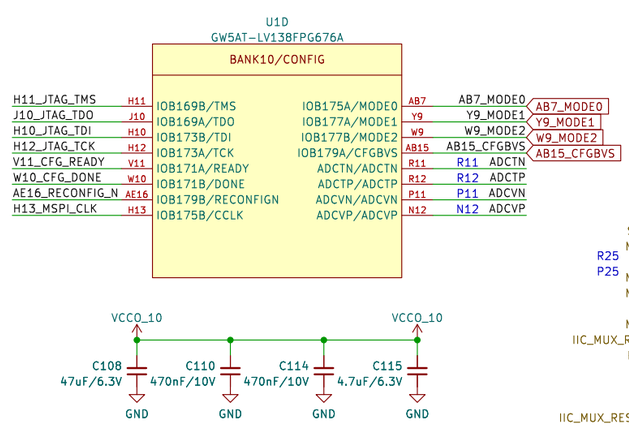
MODE SET

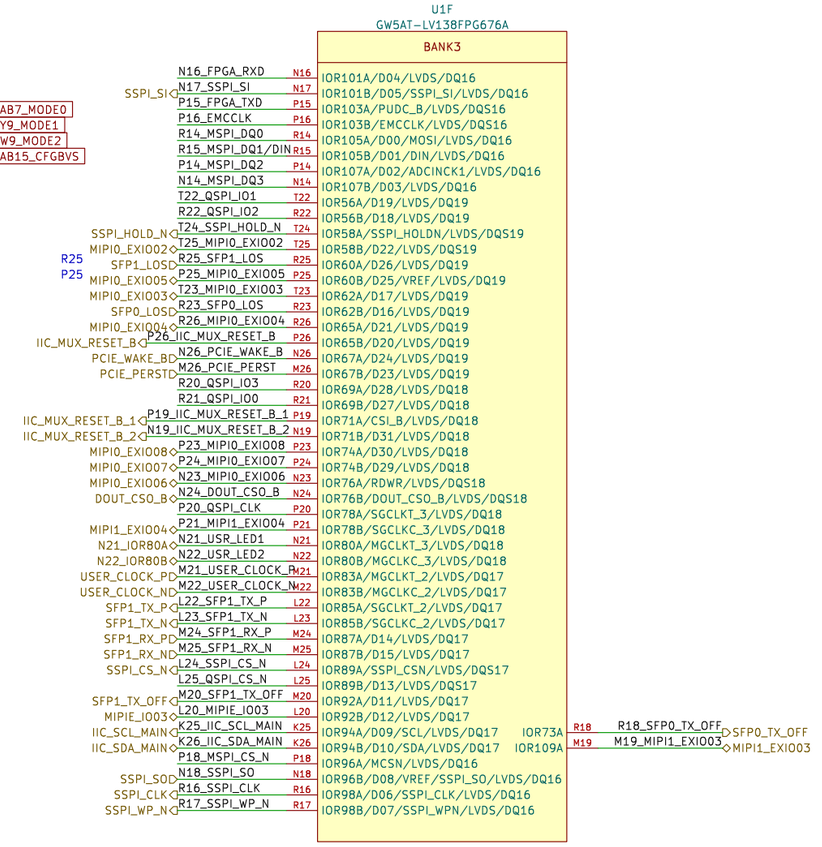


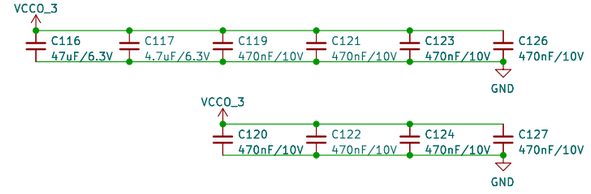
NOTE: MODE[0..2] is set to 001 by defualt.

Set CFGBVS to 1 to enable maximum driving of the pin during FPGA configuration

CONFIG	MODE2	MODE1	MODE0
JTAG	X	X	X
MSPI	0	0	1
SSPI	0	1	0
MSERIAL	0	0	0
SSERIAL	1	1	1
MCPU	1	0	0
SCPU	1	1	0







A Compact SOM for Arora V

Shenzhen Sipeed Technology Co,Ltd.

Sheet: /JTAG&CONFIG/

File: JTAG.kicad_sch

Title: TANG_MEGA-138K-Pro_4072_Schematics

Size: A3	Date: 2024-03-25	**Rev: 1.10**
KiCad E.D.A. kicad 7.0.10		Id: 7/9

DDR3 512Meg x16 0#

DDR3 512Meg x16 1#

PIN TABLE FOR DDR

DATA		MASK		CLOCK	
DDR	FPGA	DDR	FPGA	DDR	FPGA
D0	G4	LDM0	F4	CKP	M2
D1	J6	UDM0	H9	CKN	L2
D2	L8	LDM1	E3		
D3	G5	UDM1	A3		
D4	K7				
D5	J5	DQS			
D6	K8	DDR	FPGA		
D7	K6	SL0P	J4		
D8	E6	SL0N	H4		
D9	H8	SU0P	H7		
D10	H6	SU0N	G7		
D11	G8				
D12	D6	SL1P	B5		
D13	F8	SL1N	A5		
D14	G6	SL1P	C1		
D15	F7	SL1N	B1		
D16	C4	CTRL			
D17	F3	DDR	FPGA		
D18	B4	ODT	J1		
D19	E5	CKE	L3		
D20	D3	WE	J3		
D21	D5	CAS	H1		
D22	A4	RAS	H2		
D23	D4	CS	L4		
D24	E1	RST	N8		
D25	A2	BANK			
D26	G2	DDR	FPGA		
D27	C2	BA0	M4		
D28	F2	BA1	L5		
D29	E2	BA2	K3		
D30	G1				
D31	D1				

ADDR	
DDR	FPGA
A0	N1
A1	R1
A2	R2
A3	N2
A4	P1
A5	T2
A6	N4
A7	U1
A8	T4
A9	T3
A10	M1
A11	P4
A12	N3
A13	U2
A14	U5
A15	M6

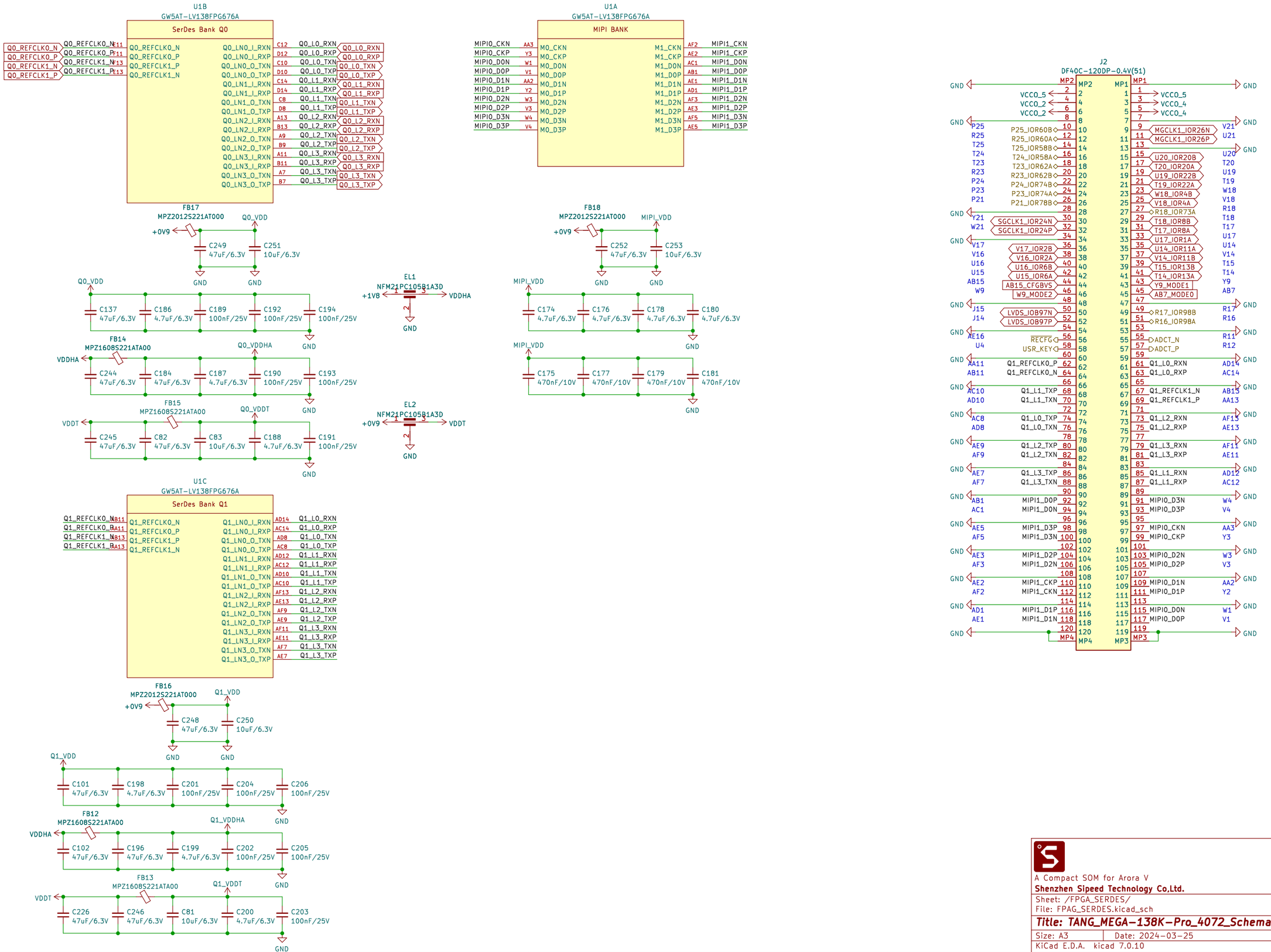
DDR Power For VTT/VREF

A Compact SOM for Arora V
Shenzhen Sipeed Technology Co,Ltd.

Sheet: /DDR3/
File: DDR3.kicad_sch

Title: TANG_MEGA-138K-Pro_4072_Schematics

Size: A3	Date: 2024-03-25	**Rev: 1.10**
KiCad E.D.A. kicad 7.0.10		Id: 8/9


U1B
GW5AT-LV138FPG676A
SerDes Bank Q0
Q0_REFCLK0_N
Q0_REFCLK0_P
Q0_REFCLK1_N
Q0_REFCLK1_P
Q0_LN0_I_RXN
Q0_LN0_I_RXP
Q0_LN0_O_TXN
Q0_LN0_O_TXP
Q0_LN1_I_RXN
Q0_LN1_I_RXP
Q0_LN1_O_TXN
Q0_LN1_O_TXP
Q0_LN2_I_RXN
Q0_LN2_I_RXP
Q0_LN2_O_TXN
Q0_LN2_O_TXP
Q0_LN3_I_RXN
Q0_LN3_I_RXP
Q0_LN3_O_TXN
Q0_LN3_O_TXP
U1A
GW5AT-LV138FPG676A
MIPI BANK
MIPI0_CKN
MIPI0_CKP
MIPI0_D0N
MIPI0_D0P
MIPI0_D1N
MIPI0_D1P
MIPI0_D2N
MIPI0_D2P
MIPI0_D3N
MIPI0_D3P
M0_CKN
M0_CKP
M0_D0N
M0_D0P
M0_D1N
M0_D1P
M0_D2N
M0_D2P
M0_D3N
M0_D3P
M1_CKN
M1_CKP
M1_D0N
M1_D0P
M1_D1N
M1_D1P
M1_D2N
M1_D2P
M1_D3N
M1_D3P
MIPI1_CKN
MIPI1_CKP
MIPI1_D0N
MIPI1_D0P
MIPI1_D1N
MIPI1_D1P
MIPI1_D2N
MIPI1_D2P
MIPI1_D3N
MIPI1_D3P
J2
DF40C-120DP-0.4V(51)
VCCO_5
VCCO_2
VCCO_4
MGCLK1_IOR26N
MGCLK1_IOR26P
U20_IOR20B
T20_IOR20A
U19_IOR22B
T19_IOR22A
W18_IOR4B
V18_IOR4A
R18_IOR73A
T18_IOR8B
T17_IOR8A
U17_IOR1A
U14_IOR11A
V14_IOR11B
T15_IOR13B
T14_IOR13A
Y9_MODE1
AB7_MODE0
R17_IOR98B
R16_IOR98A
ADCT_N
ADCT_P
V17_IOR2B
V16_IOR2A
U16_IOR6B
U15_IOR6A
AB15_CFGBVS
W9_MODE2
LVDS_IOB97N
LVDS_IOB97P
RECFG
USR_KEY
Q1_REFCLK0_P
Q1_REFCLK0_N
Q1_REFCLK1_N
Q1_REFCLK1_P
Q1_L0_RXN
Q1_L0_RXP
Q1_L1_TXP
Q1_L1_TXN
Q1_L0_TXP
Q1_L0_TXN
Q1_L2_TXP
Q1_L2_TXN
Q1_L3_TXP
Q1_L3_TXN
Q1_L2_RXN
Q1_L2_RXP
Q1_L3_RXN
Q1_L3_RXP
Q1_L1_RXN
Q1_L1_RXP
MIPI1_D0P
MIPI1_D0N
MIPI1_D3P
MIPI1_D3N
MIPI1_D2P
MIPI1_D2N
MIPI1_CKP
MIPI1_CKN
MIPI1_D1P
MIPI1_D1N
MIPI0_D3N
MIPI0_D3P
MIPI0_CKN
MIPI0_CKP
MIPI0_D2N
MIPI0_D2P
MIPI0_D1N
MIPI0_D1P
MIPI0_D0N
MIPI0_D0P
FB17
MPZ2012S221AT000
Q0_VDD
+0V9
C249
47uF/6.3V
C251
10uF/6.3V
GND
FB18
MPZ2012S221AT000
MIPI_VDD
C252
47uF/6.3V
C253
10uF/6.3V
EL1
NFM21PC105B1A3D
+1V8
VDDHA
EL2
NFM21PC105B1A3D
VDDT
C137
47uF/6.3V
C186
4.7uF/6.3V
C189
100nF/25V
C192
100nF/25V
C194
100nF/25V
FB14
MPZ1608S221ATA00
Q0_VDDHA
C244
47uF/6.3V
C184
47uF/6.3V
C187
4.7uF/6.3V
C190
100nF/25V
C193
100nF/25V
FB15
MPZ1608S221ATA00
Q0_VDDT
C245
47uF/6.3V
C82
47uF/6.3V
C83
10uF/6.3V
C188
4.7uF/6.3V
C191
100nF/25V
C174
4.7uF/6.3V
C176
4.7uF/6.3V
C178
4.7uF/6.3V
C180
4.7uF/6.3V
C175
470nF/10V
C177
470nF/10V
C179
470nF/10V
C181
470nF/10V
U1C
GW5AT-LV138FPG676A
SerDes Bank Q1
Q1_REFCLK0_N
Q1_REFCLK0_P
Q1_REFCLK1_N
Q1_REFCLK1_P
Q1_LN0_I_RXN
Q1_LN0_I_RXP
Q1_LN0_O_TXN
Q1_LN0_O_TXP
Q1_LN1_I_RXN
Q1_LN1_I_RXP
Q1_LN1_O_TXN
Q1_LN1_O_TXP
Q1_LN2_I_RXN
Q1_LN2_I_RXP
Q1_LN2_O_TXN
Q1_LN2_O_TXP
Q1_LN3_I_RXN
Q1_LN3_I_RXP
Q1_LN3_O_TXN
Q1_LN3_O_TXP
FB16
MPZ2012S221AT000
Q1_VDD
+0V9
C248
47uF/6.3V
C250
10uF/6.3V
C101
47uF/6.3V
C198
4.7uF/6.3V
C201
100nF/25V
C204
100nF/25V
C206
100nF/25V
FB12
MPZ1608S221ATA00
Q1_VDDHA
VDDHA
C102
47uF/6.3V
C196
47uF/6.3V
C199
4.7uF/6.3V
C202
100nF/25V
C205
100nF/25V
FB13
MPZ1608S221ATA00
Q1_VDDT
VDDT
C226
47uF/6.3V
C246
47uF/6.3V
C81
10uF/6.3V
C200
4.7uF/6.3V
C203
100nF/25V
GND
A Compact SOM for Arora V
Shenzhen Sipeed Technology Co,Ltd.
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